FAD

ALA-3

PERSISTENT DATA MANAGEMENT WITH DOCUMENTATION

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Class – G

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Github Repository Link:

https://github.com/shreyash2256/flutter_ala_3

Objective:

To develop a Flutter application that stores, retrieves, and updates data locally using SharedPreferences.

Workflow:

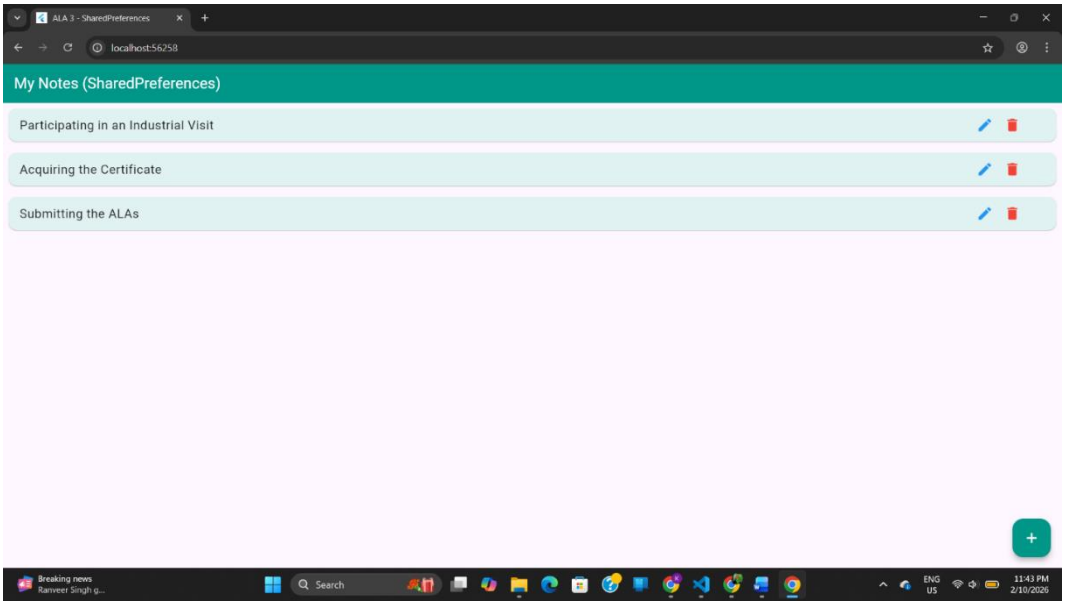
1. App starts and loads saved notes from local storage.
2. User clicks "+" to add a note.
3. Data is encoded to JSON and saved to SharedPreferences.
4. User can Edit or Delete notes, which updates the local storage immediately.

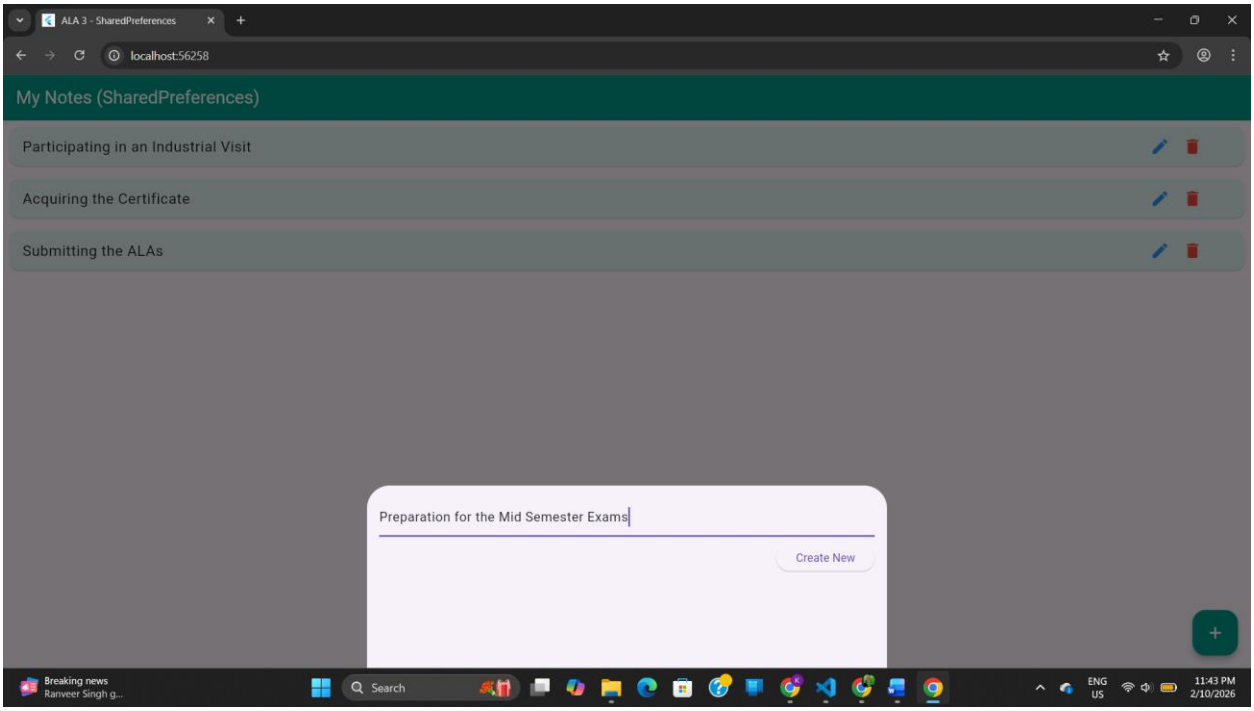
Code Snippet:

```
class StorageHelper {
  static const String _key = 'user_notes';

  // Save List of Notes to Local Storage
  static Future<void> saveNotes(List<Map<String, dynamic>> notes) async {
    final prefs = await SharedPreferences.getInstance();
    // Convert the List into a String (JSON) so SharedPreferences can store it
    final String encodedData = json.encode(notes);
```

```
    await prefs.setString(_key, encodedData);
  }

  // Load Notes from Local Storage
  static Future<List<Map<String, dynamic>>> loadNotes() async {
    final prefs = await SharedPreferences.getInstance();
    final String? encodedData = prefs.getString(_key);

    if (encodedData == null) return [];

    // Convert the String back into a List
    final List<dynamic> decodedData = json.decode(encodedData);
    return decodedData.cast<Map<String, dynamic>>();
  }
}
```

Screenshots


ALA 3 - SharedPreferences
localhost:56258
My Notes (SharedPreferences)
Participating in an Industrial Visit
Acquiring the Certificate
Submitting the ALAs
Breaking news
Ranveer Singh g...
Search
ENG
US
11:43 PM
2/10/2026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